Program: **Diploma in Civil and Environmental Engineering**	
Course Code: **5352A**	Course Title: **Sustainable Construction Practices**
Semester:**5**	Credits:**3**
Course Category: **Program Elective**	
Periods per week: **3 (L:3 T:0 P:0)**	Periods per semester: **45**

Course Objectives:

- Impart basic knowledge of sustainable construction practices
- Enable the students to use green or biomaterials in construction
- Familiarize the Sustainability Assessment Methods

Course Prerequisites:

Topic	Course code	Course name	Semester
Building Materials and Construction	3351	Building Materials and Construction	3
Sustainable Engineering	4353	Sustainable Engineering	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate knowledge of sustainability in construction	13	Understanding
CO2	Demonstrate the use of green or bio materials and Role of Building Products in sustainability	12	Understanding
CO3	Infer suitable construction techniques and practices for sustainable buildings	10	Understanding
CO4	Summarize the role of BIM and automation in sustainable construction	8	Understanding
	Series Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		3
CO2	3				3		3
CO3	3				3		3
CO4	3				3		3

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Illustrate knowledge of sustainability in construction**		
M1.01	Define Fundamental concepts of sustainability	3	Remembering
M1.02	Describe the features of sustainability indicators	2	Understanding
M1.03	Outline the concepts of sustainability analysis	1	Understanding
M1.04	Demonstrate the Environmental Impact Analysis	2	Understanding
M1.05	Explain the concept of green building	5	Understanding

Contents:

Sustainability in the built environment: sustainable development relative to ecological, economic and social conditions – efforts in sustainable development and construction – international organisations involved. Ethics and sustainability: environmental and resource concerns – resource consumption by construction industry.

Sustainability indicators- Carbon foot print, Embodied energy and water foot print, Sustainability analysis - Life Cycle Analysis.

Environmental Impact Analysis-Concept, Objectives, Environmental Impact Assessment (EIA) Process, Environmental Impact notifications in India.

Concept of Green Buildings -Green Building movement-Green buildings in India-Green building materials-Green building benefits– standards (LEED, GRIHA) – assessment structure and process. Green building rating systems – Guidelines from IGBC – LEED rating system, TERI-GRIHA rating system. Green Building Case studies – Residential, Institutional, and Commercial. Concept of Net Zero buildings – Use of BIPV and other renewable energy in buildings.

CO2	**Demonstrate the use of green or bio materials and Role of Building Products in sustainability**		
M2.01	Explain sustainable building materials	2	Understanding
M2.02	Identify the contemporary building materials	2	Remembering
M2.03	Outline the Bio materials	2	Remembering
M2.04	Summarize the use of waste materials, post-consumer materials and industrial waste	3	Understanding
M2.05	Categorize alternative building materials	3	Understanding
	Series Test I	1	

Contents:

Sustainable building materials: Introduction to sustainable building materials, qualities, use, examples - Natural building materials, locally available and locally manufactured materials – wood, earth, stone and lime-based materials.

Contemporary Building Materials- concrete, eco block, stabilized blocks (mud blocks, steam cured blocks, Fal-G Blocks stone masonry block.), insulated concrete forms(ISF), hydra form, prefabs / structural insulating panels, cellulose insulation, adobe, rammed earth, earth sheltered and recycled materials.

Bio materials: Properties, application, specification and standards (Indian and International) - Bio materials from industrial waste, mining waste, mineral waste, agricultural waste - Nontoxic materials: low VOC paints, coating and adhesives.

Use of waste materials such as paper, glass bottles, tires, shipping containers - Use of post-consumer and industrial waste such as fly-ash, bags, building construction &demolition waste – use of salvaged and recycled materials from flooring, columns, beams, timber, glass, etc.

Alternative Building Materials - Overview and definition of alternative or appropriate building materials - Alternative materials developed and promoted by government organisations like CSIR labs: CBRI and SERC, GRIHA, ASTRA (IISc), BMTPC, HUDCO and its building centres- Alternative materials developed and promoted by non-government organisations Development Alternatives (DA), Auroville, TERI.

CO3	**Identify suitable construction techniques and practices for sustainable buildings**		
M3.01	Comment on sustainable methods and technologies	2	Understanding
M3.02	Explain Ferro cement and Ferro concrete constructions	2	Understanding
M3.03	Infer Pre-engineered and ready to use building elements	3	Understanding

M3.04	Relate the contributions of agencies-costford, Nirmithi Kendra	3	Understanding

Contents:

Sustainable methods & technologies–Eco friendly and low cost techniques - Different substitute for wall construction - Flemish Bond - Rat Trap Bond. Arches – Panels - Cavity Wall .

Ferro Cement and Ferro Concrete constructions – different pre-cast members using these materials - Alternate roofing systems - Filler Slab - Composite Beam and Panel Roof.

Pre-engineered and ready to use building elements - wood products -steel and plastic –Mivan technique.

Contributions of agencies - Costford - Nirmithi Kendra – Habitat.

CO4	**Summarize the role of BIM and automation in sustainable construction**		
M4.01	Describe ICT for sustainable construction – Introduction to BIM	1	Understanding
M4.02	Define BIM for construction scheduling	3	Understanding
M4.03	Explain BIM for construction management and cost estimation	2	Understanding
M4.04	Outline Building Automation concepts	2	Understanding
	Series Test II	1	

Contents:

ICT for Sustainable Construction: Building Information modeling – Introduction to BIM, concepts and benefits.

BIM for construction scheduling, cost estimation and construction management.

Building Automation – Concepts, components of BA, applications of BA for functional efficiency of buildings

Text/Reference:

T/R	BookTitle/Author
T1	Jagadish. K.S. Alternative Building Materials and Technologies, New age International Pvt.Ltd Publishers, 2008
T2	Ross SpiegelG, Green Building Materials A Guide to Product Selection and Specification,3rd Edition by, John Wiley &Sons, 2010
T3	Traci Rose Rider, Stacy Glass, Jessica McNaughton, Understanding Green Building Materials, W.W.Norton and Company, 2011
T4	Automation Systems in Smart and Green Buildings (Modern Building Technology), Er. VK Jain, Khanna Publishers
T5	Jagadish. K.S. Building with stabilized mud,I.K. International Publishing House Pvt.Limited, 2007
R1	Sustainable Building - Design Manual Pt 1 & 2, The Energy and Resources Institute,TERI, 2004
R2	BIM Handbook: A Guide to Building Information Modeling for Owners, Managers, Designers, Engineers and Contractors- Chuck Eastman, et al.
R3	BIS, National Building Code 2005, New Delhi, 2005
R4	Energy Conservation Building Code of India, User manual, 2007
R5	P.K. Singh, Rainwater Harvesting: Low cost indigenous and innovative technologies, Macmillan Publishers India, 2008

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/
2	https://www.coursera.org/
3	https://swayam.gov.in/